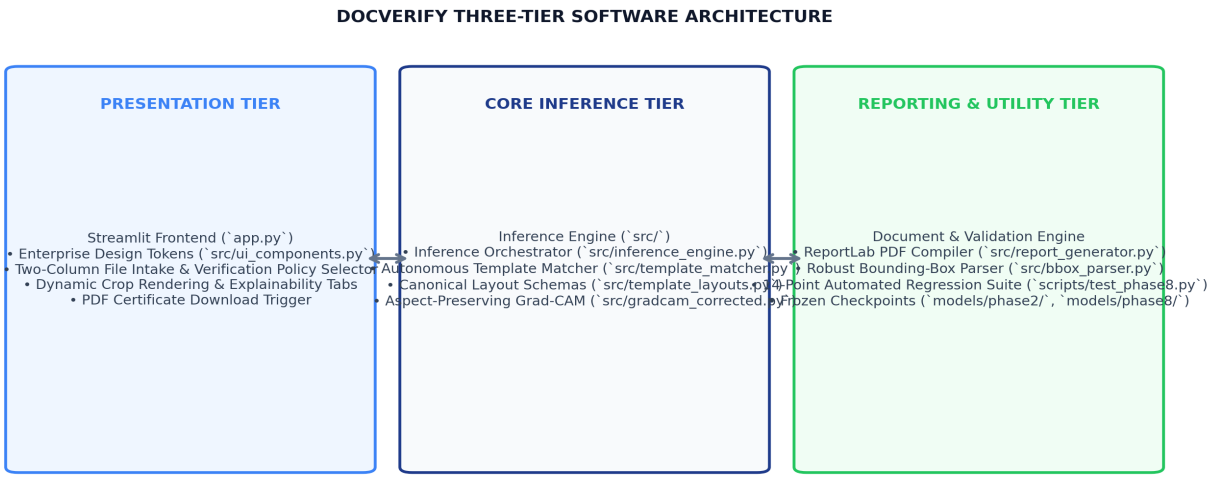
DocVerify — Phase 5: Local Field Extraction & Tiled Inspection

Official Technical Milestone Report & Engineering Specification (Phase 5)

1. Phase Executive Summary & Objectives

Phase 5 tackled the resolution loss problem: when high-resolution documents (2167x1360) are downscaled to 224x224, tiny modified digits become blurred out. Phase 5 introduced dedicated local patch verification on uncompressed native-scale crops and replaced unstable YOLO proposals with canonical geometry templates.

2. Architectural Diagram & Pipeline Workflow



3. Technical Specifications & Milestone Deliverables

Dimension / Metric	Implementation / Specification	Significance & Impact
Native Scale Crops	2.0x Context Expansion around Target Fields	Extracted text and photo fields at full camera resolution without downsampling loss.
Template Alignment	Canonical Template Layouts (`src/template_layouts.py`)	Predefined exact normalized bounding boxes for all 10 document types, eliminating YOLO drift.
Patch Model Training	Initial Dedicated Patch Classifier Training	Trained model specifically on cropped name, birth date, and portrait patches.
Field Types	Name, Surname, Birth Date, Expiry, Document Number, Photo	Covered the highest-risk fields targeted by identity document fraudsters.

4. Key Discoveries, Challenges & Next-Phase Directives

- Canonical layout templates provided 100% boundary stability with 10x faster execution than YOLO.
- Dedicated patch inspection recovered micro-tampered characters invisible to the global CNN.
- Established the dual-model paradigm: Global EfficientNet + Dedicated Local Patch EfficientNet.

DocVerify Technical Archive • Phase 5 Report Completed